

MGT-499 · RESEARCH PROJECT IN ENERGY ECONOMICS · MIDTERM ON WEDNESDAY 14 OCTOBER

Swiss Solar Project Guide

Everything for the project in one place: what each dataset is, what the literature says, a step-by-step path from raw data to slides, the presentation, and the questions to expect. Use what helps; the how-tos and answers are folded away so you can try first.

OUR QUESTION

Did Swiss communes whose household electricity got more expensive in 2023 install more rooftop solar, and is the response different where more people own their home?

Y outcome: new solar installations per commune and year

X main explanatory: 2023 tariff increase (ct/kWh)

Z heterogeneity: share of owner-occupied homes

The deliverable

- Max 8 slides, 10 min talk + 10 min Q&A
- Question and why it matters
- Ingredients: what, transformed how, limitations
- Preliminary EDA: univariate and bivariate
- Potential causality issues

How it is graded (20 points)

Slides 4 Motivatio EDA 8 Causality

The data analysis is 40% of the grade.
Regression results are not required yet.

A possible week

DAY	GOAL
Mon 5 – Tue 6	Set up, agree the question, literature notes, look at every dataset
Wed 7	Build the dataset (step 4)
Thu 8	Sample and univariate analysis (steps 5–6)

Fri 9	Bivariate analysis (step 7)
Sat 10	Causality issues and slide titles (steps 8–9)
Sun 11 – Tue 13	Slides, two timed run-throughs, Q&A practice

Possible split: one person on tariffs and plants (X, Y), one on covariates and maps (Z, controls), one on graphs, one on story, literature and slides.

WHICH DATASET IS WHICH

Datasets

All files are in `data/raw/` unless noted. **CORE** builds Y, X, Z and the panel. **OPTIONAL** adds controls or covers a factor from our brainstorm. **DON'T USE** explains why.

Core: the variables of the question

ElCom household tariffs → X

CORE

File	<code>elcom_tariffs_H2_H4_H7.csv</code>
What	Electricity price (ct/kWh) per commune × grid operator × year, 2011–2027, for household profiles H2 (2,500 kWh/yr), H4 (4,500 kWh/yr, the standard household benchmark) and H7 (13,000 kWh/yr, house with heat pump). Columns: <code>total</code> , and its parts <code>energy</code> , <code>grid</code> , <code>charge</code> , <code>aidfee</code> .
Source	Federal Electricity Commission (ElCom), via the LINDAS linked-data service · strompreis.elcom.admin.ch
How we use it	X = H4 <code>total</code> in 2023 minus 2022. Also the tariff path 2015–2027 and the energy share of the increase.
Watch out	Some communes have several operators: average them. Commune numbers are old ones: map them to 2026 (crosswalk).

SFOE plant register → Y

CORE

File	<code>bfe_plants.csv.zip</code> (extract of 14 Sep 2026)
What	Every grid-connected electricity plant: date of commissioning, capacity (kW), type (<code>SubCategory</code> : <code>subcat_2</code> = solar), mounting (<code>PlantCategory</code> : <code>plantcat_8</code> attached, <code>plantcat_9</code> integrated, <code>plantcat_10</code> free-standing), coordinates (<code>_x</code> , <code>_y</code> in Swiss LV95), postcode and locality. Catalogue files in the zip translate the codes.
Source	Swiss Federal Office of Energy (SFOE/BFE), geo.admin.ch dataset "Electricity production plants"
How we use it	Count new solar installations per commune and year (Y); capacity and size classes for detail.
Watch out	The date is grid connection, not purchase. 2026 is incomplete. 0.7% of plants have no coordinates.

Census 2000 dwellings by occupant → Z

CORE

File	bfs_dwellings_by_occupant_2000_all.csv
What	Dwellings per commune and room count, by occupant type: tenant, cooperative member, owner of a flat, sole or joint owner of the house, etc.
Source	Federal Statistical Office (FSO), federal census 2000
How we use it	Z = owner-occupied dwellings ÷ all dwellings. Split at the median (56%).
Watch out	25 years old and only per commune. 35 communes do not match after mergers.

Commune mergers (crosswalk)

CORE

File	data/processed/bfs_commune_mutations.csv
What	Every commune merger since 2010: old number, new number, date.
Source	FSO official commune register
How we use it	Step one, always: convert every commune number in every file to its 2026 number.

Commune boundaries

CORE

File	swiss_boundaries_2020.geojson
What	Commune polygons (2020) in longitude/latitude.
Source	swisstopo, swissBOUNDARIES3D
How we use it	Place each plant in a commune by its coordinates; draw maps; find neighbouring communes.
Watch out	Keep <code>OBJEKTART = "Gemeindegebiet"</code> (lakes are separate features). Map numbers to 2026.

Postcode directory

CORE

File	ortschaftenverzeichnis_plz.csv.zip
What	Postcode + locality → commune number, with the share of addresses in each commune.
Source	swisstopo
How we use it	Fallback to place the 0.7% of plants without coordinates.

Population and cantons

CORE

File	bfs_population.csv
What	Resident population per commune, 2010–2024. Cantons ("- Zürich") and districts (">> Bezirk ...") appear as header rows in the same column.
Source	FSO STATPOP
How we use it	Canton of each commune; per-1,000-inhabitant values; commune size.
Watch out	Encoded in latin1.

Controls and description

Tax income 2021

CORE

File	<code>estv_municipal_income_2021.csv</code> (cleaned from <code>estv_tax_income_2021.xlsx</code>)
What	Taxpayers and total taxable income per commune, plus income brackets.
Source	Federal Tax Administration (ESTV), direct federal tax 2021
How we use it	Mean taxable income; also the "electricity cost relative to income" factor (H4 bill ÷ income).
Watch out	Rows with numbers ≥ 9000 are totals: drop them, and drop repeated rows.

Buildings by type

CORE

File	<code>bfs_gws_buildings_2015_2024.csv</code>
What	Number of buildings per commune and year by category: <code>_T</code> all, <code>1021</code> single-family houses, <code>1025</code> multi-family.
Source	FSO building and dwelling statistics (GWS)
How we use it	Single-family share in 2021 (building suitability for rooftop solar).
Watch out	Canton rows (letters, e.g. "ZG") are mixed in: keep numeric codes. Use 2021, before the shock. <code>gws_efh_2023_2024.csv</code> has only post-shock years.

Roof solar potential

CORE

File	<code>bfe_solar_potential_communes.json</code>
What	Potential solar electricity of all roofs (and facades) per commune, GWh/yr, with monthly values.
Source	SFOE "Sonnendach" (sonnendach.ch)
How we use it	Potential per inhabitant: captures both sunlight and roof space (our "geography" and "building suitability" factors).

Votes

CORE

Files	<code>vote_20170521_communes.csv</code> (Energy Act 2017), <code>vote_20230618_communes.csv</code>
What	Yes share and turnout per commune.
Source	Federal Chancellery (voteinfo)
How we use it	Environmental attitudes, measured before the shock (2017).
Watch out	The 2023 file holds three ballots: <code>vorlagenId</code> 6620 minimum tax, 6630 Climate Act, 6640 COVID Act.

Commune typologies

CORE

File	<code>bfs_commune_levels_2025.csv</code>
What	Urban/rural type, language region, mountain area, district, agglomeration.
Source	FSO commune register

How we use it Describe the sample; compare like with like.

Grid operators

CORE

File `elcom_operators.csv`

What Operator number → name.

How we use it Label operators (who had the biggest increases); operator is also the group for clustering later.

Optional: covering the factors from our brainstorm

Energie Reporter

OPTIONAL

File `energyreporter_historized.zip` (national, canton and municipality files + README)

What Monthly indicators per commune since 2021: share of electric cars, share of solar potential already used, share of renewable heating, electricity consumption per inhabitant (where available).

Source SFOE / EnergieSchweiz "Energie Reporter"

Use for Factor 6 (consumption and lifestyle); "how much solar was already there".

Electric vehicles

OPTIONAL

Files `swisscarinfo_ev_muni_history.json`, `swisscarinfo_ev_muni_current.json`

What Cars, electric and plug-in hybrid cars per commune, yearly snapshots from 2022.

Use for Factor 6: households with an EV may value solar more. Starts in 2022, so only one pre-shock point.

Tourism (hotel nights)

OPTIONAL

Files `hesta_communes_monthly.csv`, `hesta_commune_codes.csv`

What Hotel arrivals and overnight stays for about 185 tourist communes (FSO HESTA).

Use for Factor 7 (property use, tourism). Identified by name: use the codes file to get numbers.

Weather stations

OPTIONAL

Files `meteo_nime_monthly.csv`, `meteo_nime_stations.csv`

What MeteoSwiss manual stations: monthly precipitation (`rre150m0`) and snow, with station coordinates.

Use for Factor 1 (climate) via the nearest station. No sunshine here; roof potential already includes sunlight, and comparing within cantons removes most regional weather.

Google Trends

OPTIONAL

File	<code>google_trends_swiss_solar.csv</code>
What	Monthly Swiss search interest for "Photovoltaik", "Solaranlage" and "Strompreis", 2017–2024 (index 0–100).
Use for	A motivation graph: attention to prices and solar peaked in 2022. National only.

Solar installers

OPTIONAL

Files	<code>swissolar_all.json</code> , <code>swissolar_profi.json</code> , <code>swissolar_raw.html</code>
What	Swissolar directory of solar firms with addresses and coordinates.
Use for	Factor 3 (installation capacity): installers near each commune. Today's list, not 2022's: firms founded after the boom are included.

Businesses and jobs

OPTIONAL

File	<code>bfs_statent_muni_2020_2021.json</code> (JSON-stat format)
What	Establishments and employment per commune and economic sector, 2020–2021 (FSO STATENT).
Use for	Economic structure of the commune.

Heating systems

OPTIONAL

Files	<code>bfs_heating_asset_36827394.xlsx</code> (buildings), <code>bfs_heating_asset_36827611.xlsx</code> (dwellings)
What	Heating system and energy source by canton, 2025.
Use for	Context only: canton level and after the shock.

Don't use (or not yet)

VESE feed-in tariffs

LICENCE FIRST

Files	<code>pvtarif_interface_opendata.pdf</code> (API documentation), <code>vese_municipality-supplier.csv</code>
What	What each operator pays for solar power fed into the grid.
Rule	Only through the official API with a key, after signing the free licence (info@vese.ch). Scientific use is allowed; cite "Données des tarifs de rétribution: VESE / pvtarif.ch"; never share the raw data or the key. An old decrypted copy exists in <code>archive/quarantine_vese_unlicensed/</code> : do not use it.

Old pre-built panel

DON'T USE

File	<code>solar_panel_2015_2025.csv</code>
Why	A panel from the first draft of the project. Build your own (step 4) so you know every choice in it.

Not downloaded uet

OPTIONAL

Operator areas	Grid-operator service-area polygons (geodienste.ch). Would assign operators below commune level; ElCom's commune-operator list is enough for now.
Sunshine maps	MeteoSwiss climate maps and the World Bank/Solargis PV potential map (in our brainstorm). Good pictures for the geography discussion; Sonnendach already gives a commune-level number.

WHAT WE ALREADY KNOW

Literature review

Our brainstorm's three papers, plus the closest study (Norway) and a few supporting ones. On the slides this becomes one strip on slide 2.

P01 · SWITZERLAND

Feger, Pavanini & Radulescu (2022)

Welfare and Redistribution in Residential Electricity Markets with Solar Power. *Review of Economic Studies* 89(6), 3267–3302. [Link](#)

Studies	Residential electricity pricing and solar investment in Switzerland (Bern), 2008–2014.
Finds	Solar adoption responds only weakly to changes in grid prices and subsidies.
Why it matters	The most direct Swiss evidence: prices enter the decision, but a higher price does not automatically mean a large response.
What's left	Earlier data; it does not look at the large, uneven increases of 2023.

P02 · JAPAN

Kiso, Chan & Arino (2022)

Contrasting Effects of Electricity Prices on Retrofit and New-Build Installations of Solar PV: Fukushima as a Natural Experiment. *Journal of Environmental Economics and Management* 115. [Link](#)

Studies	How price changes after Fukushima affected residential solar in Japan.
Finds	Higher prices raised installations on existing homes (retrofits); no significant effect for new builds.
Lesson	The response depends on the housing segment: look at housing composition and new construction.
Our caveat	We verified the final article from its abstract; an earlier full-text version uses a different specification, so don't mix numbers across versions.

P03 · SWEDEN

Falk & Hagsten (2025)

The Impact of Rising Electricity Prices on Demand for Photovoltaic Solar Systems. *Energy Economics* 147. [Link](#)

Studies	Whether the regional price split in Sweden after 2021 was followed by different solar growth across municipalities.
Finds	Municipalities in higher-price regions grew faster, but price differences explain less than half of the growth; much of it is common to all regions.
Lesson	Even if we find a positive link, don't attribute the whole solar boom to prices.
Our caveat	Full text not available to us: the exact price measure and whether the outcome is residential-only are unverified.

P04 · NORWAY · CLOSEST TO US

Filippini & Garnache (working paper)

Projection Bias in Durable Goods Investment: Evidence from Solar Panel Adoption in Norway.

[Seminar page](#)

Studies	The same 2022–23 crisis in Norway, where prices differ between price zones; compares treated and control regions, including near the zone borders.
Finds	The price shock raised installations strongly, and households seem to have put too much weight on crisis prices when judging long-run profitability.
Lesson	A crisis response may be partly temporary or exaggerated: careful with extrapolation.

SUPPORTING

Background and methods

- Gillingham & Tsvetanov (2019), *Quantitative Economics* 10(1): demand for solar in Connecticut; price elasticity of about -0.65 with respect to system prices.
- De Groote & Verboven (2019), *American Economic Review* 109(6): Belgian households discounted future solar benefits heavily; up-front subsidies worked better than future payments.
- Borenstein (2017), *JAERE* 4(S1): how tariff design and subsidies shape the private return to residential solar.
- Chen & Roth (2024), *Quarterly Journal of Economics* 139(2): why $\log(1+y)$ is problematic with small counts and zeros (relevant for Y).

Overall takeaway and our gap

Higher electricity prices may encourage solar adoption, but the effect is neither automatic nor uniform: it depends on housing and other conditions, and broad trends explain much of the growth. We have not found a study that tests whether the uneven

2023 tariff increases across Swiss communes were followed by more rooftop solar. So we do not try to prove that "higher prices always cause more solar"; we test whether the link appears in the recent Swiss case, how strong it is, and what else matters.

FROM OUR BRAINSTORM

Other factors and how we handle them

Each factor could affect solar adoption. What matters for causality is whether it also moves with X. Comparing communes within the same canton and over time removes anything that is fixed per commune or shared by a canton in a given year.

FACTOR	COULD IT BIAS THE X → Y LINK?	HOW WE HANDLE IT	DATA
1. Geography and climate	Only if sunny places also got bigger increases	Compare within cantons; roof potential as a control	Sonnendach; weather stations (optional)
2. Local incentives	Yes, if a municipal utility raised prices and also paid subsidies	Cantonal ones: within-canton comparison. Municipal ones: not observed, a limitation	—
3. Installation costs	National prices hit everyone; installer shortages could move firms to high-X communes	Year effects; installer proximity as a check	Swissolar (optional)
4. Permits and regulation	Rules are cantonal	Within-canton comparison	—
5. Building suitability	Yes, if houses differ with X	Single-family share, roof potential	GWS, Sonnendach
6. Consumption and lifestyle	Big users gain more from solar	H7 tariff (heat-pump house) as a check; EV share, consumption	ElCom H7, Energie Reporter, EV data
7. Property use (owner, tenant, holiday rental)	This is Z: we study it directly	Owner share split; tourism as a check	Census 2000, HESTA (optional)
8. Electricity cost relative to income	Poorer communes feel the same increase more	Income control; bill ÷ income (4,500 kWh × tariff ÷ mean income)	ElCom, ESTV

FROM RAW DATA TO SLIDES

Step by step

The course brief's five steps, broken into tasks. Ticks are saved only in your own browser.

Optional fold-outs: [How to do it](#) [Our answer](#) All code was run on our data and gives these answers.

[Open all how-tos](#)[Open all answers](#)[Close all](#)

1 Get set up

- ☐ One shared folder (or Git repo) with `data/raw` and `data/processed`
- ☐ Everyone can run Python with pandas, numpy, matplotlib, shapely (and geopandas for maps)
- ☐ Decide who looks after which part, and one weekly check-in

[+ How to do it](#)

2 Question, motivation, literature (brief step 1)

- ☐ Write the question in one sentence that names Y, X and Z
- ☐ Three reasons it matters, in your own words
- ☐ Literature notes: what each paper studies, finds, and what it means for us

[+ How to do it](#)[+ Our answer](#)

3 Know your ingredients (brief step 1)

- ☐ Open every file you need; read its documentation (links in [Datasets](#))
- ☐ Data dictionary: variable, role, source, unit, type, one limitation

[+ How to do it](#)[+ Our answer](#)

4 Build the dataset: clean and combine (brief step 2)

The longest step. Do the sub-steps in order; each ends with a number to check.

- ☐ 4.0 Check what data exist for each variable

The brief asks you to explore availability before choosing the sample.

[+ Our answer](#)

☐ **4.1 Commune crosswalk (old numbers → 2026 numbers)**

Communes merged over the years; every file must use the same numbers before joining.

[+ How to do it](#)

☐ **4.2 Population and canton of each commune**

[+ How to do it](#)

☐ **4.3 Tariffs → X**

[+ How to do it](#)

[+ Our answer](#)

☐ **4.4 Solar plants → raw Y**

[+ How to do it](#)

[+ Our answer](#)

☐ **4.5 Place each plant in a commune**

The register's locality text is not a commune number; coordinates are the most reliable.

[+ How to do it](#)

[+ Our answer](#)

☐ **4.6 Commune × year panel, 2015–2025**

[+ How to do it](#)

[+ Our answer](#)

☐ 4.7 Owner-occupier share → Z[+ How to do it](#)[+ Our answer](#)☐ 4.8 Controls: income, single-family houses, 2017 vote[+ How to do it](#)[+ Our answer](#)☐ 4.9 Sanity checks and save[+ How to do it](#)**5****Describe the sample** (brief step 2)

- ☐ Write which years and communes you keep, and why
- ☐ Summary table: N, mean, SD, min, median, max
- ☐ Map of X by commune

[+ How to do it](#)[+ Our answer](#)**6****One variable at a time** (brief step 3)

For each variable: its type, its shape, outliers, behaviour over time and space, and whether to transform it.

- ☐ Y: national yearly total, distribution per commune-year, zeros, transformation choice
- ☐ X: histogram, extreme values, spread inside each canton, map
- ☐ Z: histogram and where you split it

[+ How to do it](#)[+ Our answer](#)**7****Two variables together** (brief step 4)

- ☐ Pick the tool that fits the variable types
- ☐ Low / middle / high X: installations per year (2021 = 100)
- ☐ Growth in Y against X within cantons (binned scatter)
- ☐ The same comparison for high-Z and low-Z communes
- ☐ Correlation table: X with commune characteristics

[+ How to do it](#)

[+ Our answer](#)

8

Taste the recipe: problems and causality (brief step 5)

- ☐ List the problems you faced with the data
- ☐ Is there something else? (omitted variables)
- ☐ Is it the reverse? (reverse causality)
- ☐ Can we extrapolate? (sample bias, external validity)
- ☐ One line on how the final paper deals with each

[+ How to do it](#)

[+ Our answer](#)

9

Slides and practice (deliverable)

- ☐ At most 8 slides; each title says what the slide shows
- ☐ Every chart: axis labels with units, a source line, readable from the back, one y-axis
- ☐ Two timed run-throughs under 10 minutes
- ☐ Each person prepares two likely questions on their part (see [Q&A](#))

[+ How to do it](#)

8 SLIDES · ABOUT 75 SECONDS EACH · PLUS 4 BACKUPS

Presentation

The actual slides, with what to say under each one. Click a slide to view it full screen; use the arrow keys to move and Esc to close. The editable deck is the separate page "Swiss Solar Midterm Deck".

Present from slide 1

Open all speaker notes

Slide 1

Did the 2023 electricity price shock speed up rooftop solar in Switzerland?

Motivation

+ Speaker notes

Slide 2 Why it matters, and what we already know

Motivation

+ Speaker notes

Slide 3 Y, X and Z come from official open data

EDA · Clarity

+ Speaker notes

Slide 4 Installations rose with tariffs in 2023, then fell back

EDA

+ Speaker notes

Slide 5 The shock was large, uneven, and mostly within cantons

EDA

+ Speaker notes

Slide 6 High-shock communes installed more after 2023

EDA

+ Speaker notes

Slide 7 Larger increases, faster growth, in both kinds of commune

EDA · Heterogeneity

+ Speaker notes

Slide 8 Is the link causal? Three questions

Causality

+ [Speaker notes](#)

+ Backup slides for questions (not presented)

TEN MINUTES OF QUESTIONS

Q&A preparation

- + "Isn't the tariff increase related to rich or green communes?"
- + "Isn't this the feed-in tariff, not the retail price?"
- + "Why not just use log(installations)?"
- + "Did installers just move to high-tariff communes?"
- + "Why owner share from 2000?"
- + "Is the effect permanent?"
- + "Why Switzerland and not the whole of Europe?"
- + "What did you exclude, and could that bias the result?"

NOTHING FORGOTTEN

Brief check

Every point of the course brief, and where this guide covers it.

BRIEF	WHERE
Causal question with at least two variables, ideally outcome + explanatory + heterogeneity	Overview; step 2; slide 1
Literature search on the state of research	Literature; step 2; slide 2
Explore the datasets, read the documentation	Datasets; step 3
Explore data availability	Step 4.0

Clean the data and select a clean sample	Steps 4.1–4.9 and 5
Describe the sample: summary tables, graphs, maps	Step 5; slides 4–5
Univariate: nature of each variable, transformations, outliers, time and space	Step 6; slides 4–5
Bivariate: right tool for the variable types, describe findings	Step 7; slides 6–7